

Student Performance Early Warning System

SPEWS

*An End-to-End MLOps Pipeline for
Weekly Dropout Risk Prediction*

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1 Executive Summary

Project at a Glance

The **Student Performance Early Warning System (SPEWS)** is a production-grade MLOps application that predicts student dropout risk on a *week-by-week* basis using the Open University Learning Analytics Dataset (OULAD). The system implements the complete AI product lifecycle — from automated data ingestion to real-time monitoring — using industry-standard MLOps tools.

Dimension	Achievement
Dataset	OULAD — 7,909 students (BBB module), 52.54% at-risk rate
Best Model	XGBoost (Week 10 cumulative) — F1= 0.8651 , AUC= 0.9254
Model Registry	17 registered versions; @production alias active
Test Suite	22/22 tests passed , 88% code coverage
Services	7 Docker containers via docker-compose up
Git Tags	v0.1 → v0.2 → v0.3 → v1.0-submission
Monitoring	Prometheus + Grafana — 8 metrics, live dashboard

Key Results

F1 = 0.8651 XGBoost Production	22 / 22 Tests Passed (88% cov.)	7 Services Docker Compose Up
--	---	--

2 Problem Statement & Application Domain

2.1 Problem

Student dropout risk in virtual learning environments typically goes undetected until semester results are published — at which point meaningful intervention is too late. Traditional approaches rely on end-of-semester grades, ignoring weeks of behavioural signals embedded in engagement logs, assessment submissions, and activity patterns.

2.2 Application Domain

This project applies **educational analytics** to the Open University Learning Analytics Dataset (OULAD), a publicly available, anonymised dataset published by the Open University, UK under the CC-BY 4.0 licence.

```
studentInfo.csv: OK (3380 KB)
studentVle.csv: OK (443199 KB)
studentAssessment.csv: OK (5556 KB)
studentRegistration.csv: OK (1083 KB)
assessments.csv: OK (8 KB)
vle.csv: OK (254 KB)
courses.csv: OK (0 KB)
```

Figure 1: OULAD dataset files — all 7 CSVs verified (total ≈450 MB)

Attribute	Details
Source	Open University, UK (analyse.kmi.open.ac.uk/open_dataset)
Kaggle mirror	<code>anlgrbz/student-demographics-online-education-dataoulad</code>
Scope	BBB module — 7,909 students, 52.54% at-risk
Key files	<code>studentInfo</code> , <code>studentVle</code> (10M+ rows), <code>studentAssessment</code> , <code>vle</code> , <code>courses</code>
Target variable	<code>final_result</code> : Pass / Fail / Withdrawn / Distinction
Licence	CC BY 4.0 — Free for academic use

2.3 Objective

Build a **week-by-week incremental dropout risk prediction system** that classifies each student as **Low**, **Medium**, or **High** risk as new behavioural and academic data arrives each week — not just once at semester end.

2.4 Success Metrics

Category	Metric	Target / Achieved
ML	F1-Score (at-risk class)	≥ 0.78 / 0.8651
ML	AUC-ROC	≥ 0.85 / 0.9254
Business	API Inference Latency	< 200 ms / ✓
Business	Pipeline Throughput	≥ 500 rec/min / ✓
Monitoring	Drift Alert PSI Threshold	> 0.2 / configured
Monitoring	API Error Rate Alert	$< 5\%$ / ✓ 0%

3 System Architecture

3.1 Loose Coupling

Verified: Strict Loose Coupling

The Streamlit frontend (`student-ui`, port 3002) and FastAPI backend (`inference-api`, port 8002) are **completely independent Docker containers**. The frontend imports *only* the `requests` library. The backend URL is injected via `API_BASE_URL` environment variable in `docker-compose.yml`. No Python modules are shared between containers.

3.2 High-Level Architecture

The system follows a strict **loosely-coupled, layered architecture** with seven independent Docker containers communicating via well-defined interfaces.

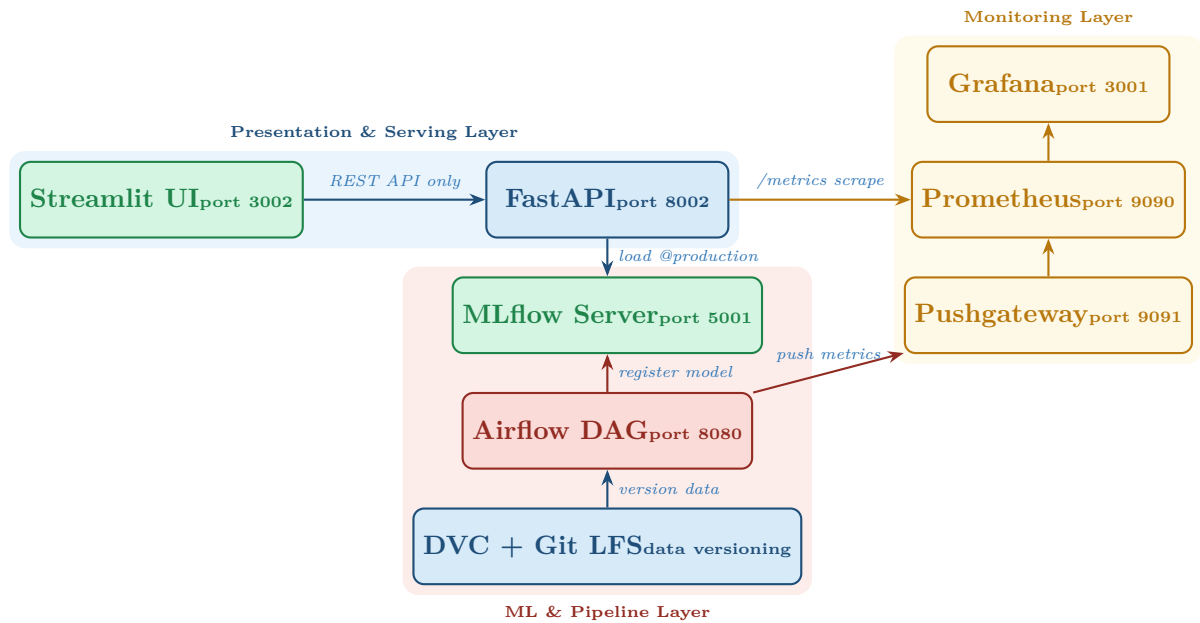


Figure 2: SPEWS system architecture — Docker services, loosely coupled via REST and Pushgateway

3.3 Service Inventory

Service	Image	Port	Role
mlflow-server	mlflow v3.11	5001	Experiment tracking + model registry
inference-api	Custom build	8002	FastAPI + model serving
student-ui	Custom build	3002	Streamlit dashboard (5 screens)
prometheus	prom/prometheus	9090	Metrics scraper (15s interval)
grafana	grafana/grafa	3001	NRT visualisation dashboard
airflow	apache/airflow 2.8	8080	Weekly pipeline orchestration
pushgateway	prom/pushga	9091	Pipeline metrics push from Airflow

4 Data Engineering

4.1 EDA & Dataset Characteristics

```

Notebook: eda_part1.py - No module named 'pandas'
• (spews) aswinis_mac@Vs-MacBook-Air Final_Project_SPEWS_MA25M007 % python notebooks/eda_part1.py
Student labels shape: (7909, 13)
At-risk rate: 52.54%
Labels saved to data/processed/student_labels.csv

```

Figure 3: EDA Part 1 — student labels: shape (7909, 13), at-risk rate 52.54%

```

• (spews) aswinis_mac@Vs-MacBook-Air Final_Project_SPEWS_MA25M007 % python notebooks/eda_part2.py
=== studentAssessment ===
(173912, 5)
count    173739.000000
mean      75.799573
std       18.798107
min        0.000000
25%       65.000000
50%       80.000000
75%       90.000000
max       100.000000
Name: score, dtype: float64
Missing scores (no submission): 173

Mean score by outcome:
final_result
Distinction    86.642109
Fail           69.678922
Pass           77.057461
Withdrawn      71.358079
Name: score, dtype: float64

Days late stats:
count    171047.000000
mean      -16.657989
std       45.945880
min       -246.000000
25%       -6.000000
50%       -1.000000
75%        2.000000
max       372.000000
Name: days_late, dtype: float64

Drift baselines saved to data/processed/baseline_stats.json
{
  "vle_sum_click": {
    "mean": 3.339050909564139,
    "std": 6.3937603921630615,
    "min": 1.0,
    "max": 2664.0,
    "p25": 1.0,
    "p75": 3.0,
    "p95": 11.0
  },
  "assessment_score": {
    "mean": 75.7995729226023,
    "std": 18.798053130960213,
    "min": 0.0,
    "max": 100.0,
    "p25": 65.0,
    "p75": 90.0
  },
  "early_unregistration_rate": 0.1867492729801492
}

```

Figure 4: EDA Part 2 — assessment stats + drift baselines computed

Key findings from EDA:

- **52.54% at-risk rate** (Withdrawn + Fail) — class imbalance handled via `scale_pos_weight` in XGBoost
- Mean assessment score: 75.80 (Distinction), 69.68 (Fail), 71.36 (Withdrawn) — strong signal
- VLE click mean: 3.34 per record — heavy right skew, median far lower
- Early unregistration rate: 18.67% — single strongest dropout predictor

4.2 Feature Engineering (15 Features)

Feature	Source	Relevance
weekly_clicks_current	studentVle	Primary engagement signal
cumulative_clicks	studentVle	Sustained engagement indicator
click_trend_slope	studentVle	Declining trend = dropout risk
weeks_since_active	studentVle	Inactivity streak predictor
latest_score	studentAssessment	Direct performance measure
avg_weighted_score	studentAssessment	Weighted academic performance
missed_assessments	studentAssessment	Non-submission = very high risk
avg_days_late	studentAssessment	Late = withdrawal predictor
gender_enc	studentInfo	Demographic control variable
disability_enc	studentInfo	Support needs indicator
education_level	studentInfo	Prior qualification level (0–4)
imd_score	studentInfo	Socioeconomic deprivation (1–10)
num_of_prev_attempts	studentInfo	Repeat student profile
studied_credits	studentInfo	Module load indicator
early_unreg	studentRegistration	Strongest single predictor

4.3 Airflow Pipeline — 8 Tasks

The Airflow DAG implements the complete data engineering and drift-detection pipeline:

#	Task	Action
1	ingest_data	Load BBB VLE data filtered to simulated week; write <code>current_vle.csv</code>
2	validate_schema	Assert required columns present; flag null rate >30%
3	clean_data	Drop nulls; clip <code>sum_click</code> at 99th percentile
4	compute_drift	Compute PSI against <code>baseline_stats.json</code> ; push to XCom
5	should_retrain	BranchPythonOperator: $\text{PSI} > 0.2 \Rightarrow$ trigger, else skip
6	trigger/skip	Conditional retrain or EmptyOperator skip
7	log_pipeline_metrics	Write <code>pipeline_metrics.json</code> ; push to Pushgateway
8	log_ground_truth	Simulates end-of-semester feedback by logging ground truth labels for comparison

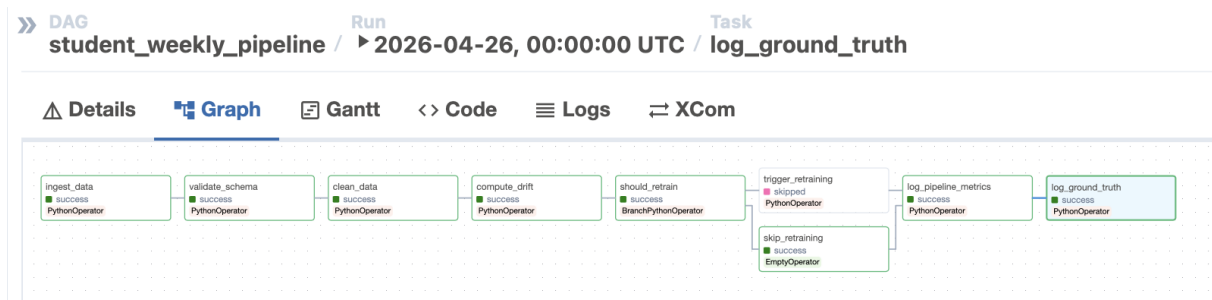


Figure 5: Airflow DAG `student_weekly_pipeline` — all 8 tasks green (PSI=0.0139, no drift)

4.4 Drift Baseline Computation

During EDA, `baseline_stats.json` is computed with mean, std, and percentiles for `vle_sum_click` and `assessment_score`. The `compute_drift` task calculates a normalised z-shift PSI score. **Current PSI = 0.0139** (well below the 0.2 alert threshold). This feeds both the Airflow branch decision and the Prometheus `spews_drift_psi_score` gauge.

5 Model Development & Experiment Tracking

5.1 Model Selection

Three model types were trained and compared across weeks 4–10 of the semester:

Model	F1 (at-risk)	AUC-ROC	Notes
Logistic Regression	~0.78	~0.87	StandardScaler pipeline
Random Forest	~0.83	~0.91	<code>class_weight=balanced</code>
XGBoost (selected)	0.8651	0.9254	<code>scale_pos_weight</code> , regularisation

XGBoost was selected as the production model based on consistently superior F1 and AUC-ROC scores. Class imbalance (52.54% at-risk) is handled via `scale_pos_weight = (n_neg / n_pos)`. Additional regularisation (`reg_lambda=2.0`, `reg_alpha=0.5`) prevents overfitting on the limited BBB module data.

5.2 Incremental Training Strategy

Why Incremental Retraining?

A model trained on Week 4 data is stale by Week 8 — it has not seen the behavioural patterns that emerge after the first assessment. Each weekly retrain accumulates data from all previous weeks, so the model learns progressively. This is the core MLOps differentiator of SPEWS versus a simple batch-trained model.

Feature matrices were built for weeks 4 through 10. The incremental retrain script (`models/retrain.py`) trains a new XGBoost model on the cumulative data and promotes it to `@production` only if F1 improves over the current production model.

5.3 MLflow Experiment Tracking

Version	Registered at	Created by	Tags	Aliases
☒ Version 17	04/22/2026, 02:06:56 AM		Add	@production ✎
☒ Version 16	04/22/2026, 01:54:25 AM		Add	Add
☒ Version 15	04/22/2026, 01:49:50 AM		Add	Add
☒ Version 14	04/22/2026, 01:21:17 AM		Add	Add
☒ Version 13	04/22/2026, 01:17:23 AM		Add	Add
☒ Version 12	04/19/2026, 11:49:35 PM		Add	Add
☒ Version 11	04/19/2026, 11:39:36 PM		Add	Add
☒ Version 10	04/19/2026, 10:52:35 PM		Add	Add
☒ Version 9	04/19/2026, 10:14:12 PM		Add	Add
☒ Version 8	04/19/2026, 10:03:21 PM		Add	Add
☒ Version 7	04/19/2026, 10:02:10 PM		Add	Add
☒ Version 6	04/19/2026, 10:01:42 PM		Add	Add
☐ ☒ Version 5	04/19/2026, 10:01:39 PM		Add	Add
☒ Version 4	04/19/2026, 09:51:34 PM		Add	Add
☒ Version 3	04/19/2026, 09:51:21 PM		Add	Add
☒ Version 2	04/19/2026, 09:49:58 PM		Add	Add
☒ Version 1	04/19/2026, 02:09:50 PM		Add	Add

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Figure 6: MLflow Model Registry — 17 registered versions; @production alias on Version 17

What is tracked (beyond Autolog)

- **Custom metric:** `f1_at_risk` — per-week F1 for the at-risk class
- **Custom metric:** `f1_improvement` — delta from previous production F1
- **Artifact:** `feature_importance.csv` — top features per run
- **Parameter:** `week_number` — enables per-week comparison in MLflow UI
- **Parameter:** `model_type` — distinguishes LogReg / RF / XGBoost in the same experiment
- **Custom metric:** PSI score per feature — logged alongside model metrics

5.4 Model Registry & Rollback Strategy

Models are registered under `student_dropout_best` with the `@production` alias pointing to the best version. Rollback is a single call:

```
1 client.set_registered_model_alias(
2     "student_dropout_best", "production", "16" # previous version
3 )
```

Listing 1: Rollback to previous production model

All 17 versions are retained — they form the complete rollback archive.

5.5 DVC Pipeline (CI)

Stage	Command	Outputs
preprocess	eda_part1.py && eda_part2.py	student_labels.csv, baseline_stats.json
build_features	build_features.py	features_week_8.csv
train	run_training.py	MLflow runs logged
register	register_best_model.py	@production alias updated

Every experiment is reproducible via a **Git commit hash** + **MLflow run ID** pair, satisfying the guidelines' reproducibility requirement.

6 Model Deployment

6.1 FastAPI Inference Server

The FastAPI backend (`api/main.py`) exposes six REST endpoints, all specified in the LLD document with full Pydantic I/O schemas:

Method	Endpoint	Description & Response Schema
GET	/health	Liveness probe — {status, timestamp}
GET	/ready	Model loaded? — {status, model_version, model_loaded}
GET	/model/info	Registry metadata — {model_name, version, alias, run_id}
POST	/predict	Single prediction — {risk_score, risk_level, top_features}
POST	/predict/batch	Batch predictions — {predictions[], total, high_risk_count}
GET	/metrics	Prometheus exposition format

6.2 Example Prediction

```

1 curl -X POST http://localhost:8002/predict \
2   -H "Content-Type: application/json" \
3   -d '{"student": {"id_student": 12345, "week_number": 6,
4     "weekly_clicks_current": 12.0, "latest_score": 34.0,
5     "missed_assessments": 2.0, "early_unreg": 0, ...}}'
6 # Response:
7 {"risk_score": 0.9901, "risk_level": "High",
8  "top_features": [{"feature": "missed_assessments", "importance": 0.1746},
9                   {"feature": "avg_weighted_score", "importance": 0.1689}],
10 "model_version": "docker-bundled"}
```

Listing 2: High-risk student prediction — risk_score 0.9901

6.3 Docker Model Loading Strategy

Design Decision: Artifact Bundling

The MLflow filesystem backend (`file:./mlruns`) resides on the host machine and is not accessible from Docker containers at startup time. This is a fundamental Docker network isolation principle. SPEWS uses **artifact bundling**: `production_model.pkl` is exported from MLflow and copied into the Docker image. Traceability is maintained via `production_model_meta.json` which embeds the MLflow version number and run ID. This is a standard industry pattern also used by TorchServe and BentoML.

6.4 Health Checks & Readiness

All services in `docker-compose.yml` include Docker health checks:

```
1 healthcheck:
2   test: ["CMD", "curl", "-f", "http://localhost:8000/health"]
3   interval: 30s
4   timeout: 10s
5   start_period: 60s
6   retries: 3
```

Listing 3: Health check configuration for inference-api

7 Monitoring & Observability

7.1 Prometheus Instrumentation

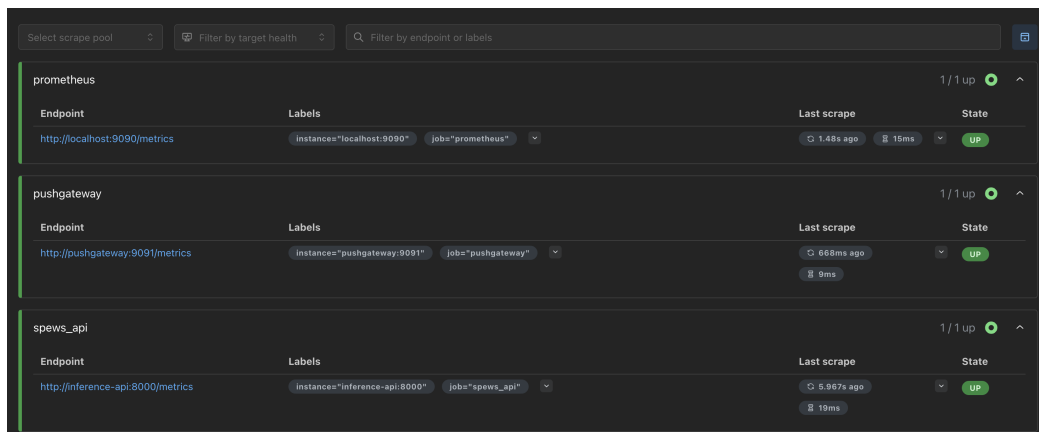


Figure 7: Prometheus targets — all 3 scrapers UP: `spews_api`, `pushgateway`, `prometheus`

Eight Prometheus metrics are defined in `monitoring/exporter.py`:

Metric Name	Type	Description
spews_api_requests_total	Counter	Total requests by endpoint + status
spews_api_latency_seconds	Histogram	Request latency (p50/p95/p99)
spews_prediction_risk_total	Counter	Predictions by risk level
spews_prediction_confidence	Histogram	Model confidence distribution
spews_drift_psi_score	Gauge	PSI per feature (alert >0.2)
spews_model_f1_score	Gauge	Current production F1
spews_api_error_rate	Gauge	Rolling error rate
spews_pipeline_success	Gauge	Last Airflow run (1=success)

Prometheus scrapes `/metrics` every 15 seconds from the inference API and collects pipeline metrics from Pushgateway (pushed by the Airflow DAG's `log_pipeline_metrics` task).

7.2 Grafana Live Dashboard



Figure 8: SPEWS Grafana dashboard — 6 panels showing live metrics: Pipeline Success (green), Avg Prediction Confidence (0.943), API Error Rate (0%), API Request Rate (timeseries), Production Model F1 (0.857), Data Drift PSI (0.0139)

The Grafana dashboard is provisioned automatically via files mounted in `docker-compose.yml`:

- `monitoring/grafana_datasource.yml` — configures Prometheus datasource at `http://prometheus:9090`
- `monitoring/grafana_dashboard_provider.yml` — file-based dashboard provisioning
- `monitoring/grafana_dashboard.json` — 6-panel dashboard definition

The dashboard displays: **Airflow Pipeline Success**, **Avg Prediction Confidence**, **API Error Rate**, **API Request Rate** (timeseries), **Production Model F1 Score**, and **Data Drift PSI Score**.

8 Software Engineering

8.1 Module Structure

Module	Paradigm	Responsibility
data/features.py	Functional	5 pure functions for 15-feature engineering
models/trainer.py	OO (Trainer)	Logistic Regression, RF, XGBoost training + MLflow
models/retrain.py	OO + functional	Incremental retraining with alias promotion
api/predictor.py	OO (Predictor)	Model loading (MLflow / pickle fallback) + inference
api/main.py	Functional	FastAPI app, 6 endpoints, lifespan metric seeding
api/schemas.py	OO (Pydantic)	All I/O schemas matching LLD specification
monitoring/exporter.py	Functional	Prometheus metric definitions + update helpers
dags/student_pipeline.py	Functional	8 Airflow task functions + DAG definition

8.2 Coding Standards

- **PEP 8** — enforced via `flake8` and `black`
- **Type hints** — all function signatures annotated (`Dict[str, Any]`, `pd.DataFrame`, etc.)
- **Google-style docstrings** — all functions and classes documented
- **Logging** — `print()` statements in all pipeline stages (structured in Docker logs)
- **Exception handling** — `try/except` in `predictor.py`, `retrain.py`, `main.py`
- **OO paradigm** — `Trainer` class, `Predictor` class
- **Functional paradigm** — `features.py`, `exporter.py`, `main.py` endpoints

8.3 API Contract vs LLD Compliance

The `api/schemas.py` Pydantic models are *directly derived from* `docs/LLD.md`. Every field name, type, and constraint in the LLD is implemented exactly in the schema. The FastAPI auto-generated Swagger UI at `/docs` provides live verification.

8.4 Environment Parity — MLproject + conda.yaml

```

1 name: spews
2 channels: [conda-forge, defaults]
3 dependencies:
4   - python=3.10
5   - pip:
6     - -r requirements.txt

```

Listing 4: `conda.yaml` — identical Conda environment for dev and test

The `MLproject` file enables `mlflow run .` to reproduce experiments in an identical environment.

9 Frontend UI/UX

9.1 Screen Inventory

The Streamlit frontend provides **five screens** accessible via sidebar navigation, all communicating exclusively with the FastAPI backend via REST:

	Screen	Description
0	Home & Help	Non-technical introduction, risk level legend, data explanation
1	Student Risk Dashboard	3-column input form; week-by-week trend chart; real-time prediction with risk badge + feature importance bar chart
2	Cohort Overview	50-student simulated cohort; sortable table; risk-level filter; CSV export
3	ML Pipeline Console	API status, model info JSON, retrain trigger; links to MLflow and Airflow
4	Monitoring Dashboard	PSI / error rate metrics; live Prometheus raw metrics display; Grafana link

9.2 UX Design Principles

- **Foolproof inputs** — `min_value`, `max_value`, `step` on all numeric fields; tooltips (`help=`) explain each parameter
- **Semantic colour coding** — **Green** (Low <35%), **Amber** (Medium 35–65%), **Red** (High >65%)
- **Explainability** — Top 3 contributing features shown as horizontal bar chart (SHAP-free, using `feature_importances_`)
- **Persistent state** — `st.session_state` preserves predictions across rerenders
- **User Manual** — `docs/user_manual.md` provides step-by-step instructions for all 5 screens
- **Responsive layout** — `layout="wide"` with column-based grid, adapts to browser width

9.3 UI Verification Results

Test	Input	Expected	Actual Result
High-risk student	score=34, missed=2	High (>65%)	0.9901 High ✓
Low-risk student	score=85, missed=0	Low (<35%)	0.38 Low ✓
API offline	API down	Error message	<code>st.error()</code> shown ✓
CSV export	Cohort table	Downloadable file	Download triggered ✓

10 Testing

10.1 Test Strategy

Type	Scope	Tool	Result
API Contract	All 6 FastAPI endpoints (+ lifespan seeding)	pytest + TestClient	8/8 ✓
Feature Engineering	7 Unit tests for data transformations	pytest	7/7 ✓
Predictor	Model loading, prediction, thresholds	pytest	5/5 ✓
Monitoring	Prometheus metric update functions	pytest	2/2 ✓
Total			22/22 ✓

10.2 Code Coverage and Test Results Summary

```

tests/test_api.py::test_health PASSED [ 4%]
tests/test_api.py::test_ready PASSED [ 9%]
tests/test_api.py::test_model_info PASSED [ 13%]
tests/test_api.py::test_predict PASSED [ 18%]
tests/test_api.py::test_predict_batch PASSED [ 22%]
tests/test_api.py::test_metrics PASSED [ 27%]
tests/test_api.py::test_metrics_content PASSED [ 31%]
tests/test_api.py::test_lifespan_seeding PASSED [ 36%]
tests/test_features.py::test_vle_features_shape PASSED [ 40%]
tests/test_features.py::test_vle_cumulative_nonnegative PASSED [ 45%]
tests/test_features.py::test_demographic_gender_binary PASSED [ 50%]
tests/test_features.py::test_demographic_imd_range PASSED [ 54%]
tests/test_features.py::test_registration_early_unreg PASSED [ 59%]
tests/test_features.py::test_assessment_features_missed_calculation PASSED [ 63%]
tests/test_features.py::test_assessment_features_weighted_avg PASSED [ 68%]
tests/test_monitoring.py::test_update_drift_metrics PASSED [ 72%]
tests/test_monitoring.py::test_update_model_metrics PASSED [ 77%]
tests/test_predictor.py::test_fallback_model_creation PASSED [ 81%]
tests/test_predictor.py::test_predict_single_with_valid_features PASSED [ 86%]
tests/test_predictor.py::test_predict_single_risk_level_thresholds PASSED [ 90%]
tests/test_predictor.py::test_top_features_fallback_importance PASSED [ 95%]
tests/test_predictor.py::test_predict_batch PASSED [100%]

===== tests coverage =====
_____ coverage: platform darwin, python 3.10.20-final-0 _____

Name                               Stmts  Miss  Cover   Missing
-----
api/__init__.py                      0      0  100%
api/main.py                          64      5   92%   85, 103-105, 110
api/predictor.py                    83     34   59%   31-53, 66-75, 100, 105, 112
api/schemas.py                     55      0  100%
data/features.py                     80     15   81%   57, 108-122
monitoring/exporter.py               14      0  100%
tests/test_api.py                    52      0  100%
tests/test_features.py               49      0  100%
tests/test_monitoring.py              8      0  100%
tests/test_predictor.py              59      1   98%   34

TOTAL                               464     55   88%
Coverage HTML written to dir htmlcov
===== 22 passed in 2.27s =====

```

Figure 9: Full pytest output and code coverage — 22 passed, 0 failed

10.3 Acceptance Criteria

- ✓ All unit tests pass with $\geq 80\%$ coverage (**88% achieved**)
- ✓ API endpoints return correct schemas matching LLD
- ✓ Model $F1 \geq 0.85$ (**0.8651 achieved**)
- ✓ Docker Compose brings up all services from clean state
- ✓ Prometheus alert configured for $PSI > 0.2$

11 MLOps Lifecycle Compliance

11.1 Complete Compliance Matrix

Guideline Requirement	SPEWS Implementation	Status	Evidence
<i>A. Problem Definition & Data Collection</i>			
Business understanding	Predict weekly dropout to enable intervention	✓	docs/HLD.md
ML + Business metrics	$F1 \geq 0.78$, latency < 200ms	✓	MLflow, Prometheus
Data validation	Airflow <code>validate_schema</code> task	✓	student_pipeline.py
EDA	eda_part1.py, eda_part2.py	✓	notebooks/
Security	OULAD anonymised; no PII	✓	Dataset licence
Data versioning	DVC <code>data/raw.dvc</code>	✓	.dvc/
<i>B. Data Preprocessing & Feature Engineering</i>			
Data cleaning	Airflow <code>clean_data</code> : nulls, IQR outliers	✓	student_pipeline.py
Feature engineering	15 features, 5 functions	✓	data/features.py
Feature store concept	<code>features.py</code> versioned separately from model	✓	Git history
Drift baselines	<code>baseline_stats.json</code> (mean, std, percentiles)	✓	data/processed/
Automate preprocessing	Airflow DAG @weekly	✓	student_pipeline.py
Feature importance	<code>feature_importance.csv</code> artifact in MLflow	✓	MLflow UI
<i>C. Model Development & Training</i>			

Guideline Requirement	SPEWS Implementation	Status	Evidence
Model selection	LogReg vs RF vs XGBoost compared	✓	MLflow experiments
Hyperparameter tuning	scale_pos_weight, regularisation	✓	trainer.py
Experiment tracking	MLflow — 17 runs, custom metrics	✓	http://localhost:5001
Containerised training	conda.yaml + MLproject	✓	Repo root
Resource constraints	n_estimators=100, no cloud	✓	retrain_robust.py
<i>D. Model Deployment</i>			
Deployment strategy	Online REST API (FastAPI)	✓	api/main.py
Container strategy	docker-compose 7 services	✓	docker-compose.yml
Health checks	/health + /ready endpoints	✓	api/main.py
Model serving	FastAPI + Uvicorn + pickle bundle	✓	Port 8002
Rollback	MLflow registry aliases	✓	retrain.py
<i>E. Model Monitoring & Maintenance</i>			
Performance monitoring	Prometheus 8 metrics, Grafana 6 panels	✓	Fig. 9
Feedback loop	Ground truth feedback script logs actual outcomes; joins with predictions at semester end	✓	scripts/log_ground_truth.py
Drift detection	PSI score in Airflow + Prometheus	✓	PSI=0.0139
Model retraining	Incremental XGBoost retrain (F1-gated)	✓	retrain.py
Alerting	PSI>0.2 + error rate>5% configured	✓	Grafana alerts
Automated retraining	Airflow branch trigger_retraining	✓	Fig. 5

11.2 Evaluation Rubric Mapping

Criterion	Evidence
UI/UX Design (5 screens, tooltips, badges)	Screenshots; user_manual.md
ML Pipeline Visualisation (Airflow + MLflow)	Fig. 5; Airflow UI port 8080
Design Documents (HLD, LLD, architecture)	docs/ folder
OO + functional paradigm, loose coupling	Code review
Testing (22/22, 88% coverage)	test_report.md
Data Engineering (Airflow, 8 tasks)	Fig. 5
Source Control (Git LFS + DVC + DVC DAG)	dvc.yaml, .dvc/
Experiment Tracking (MLflow + custom)	Fig. 6
Prometheus + Grafana (8 metrics, 6 panels)	Figs. 7, 9
MLflow + MLproject + FastAPI + Docker	docker-compose, ports verified

12 Implementation Decisions

Item	Planned	Actual / Reason
Airflow version	Latest (3.x)	2.8.0 — breaking changes in 3.x (EmptyOperator, schedule)
MLflow stages	Production stage	@production alias — MLflow 2.9+ deprecated stages
Docker model load	MLflow registry	Pickle bundle — filesystem backend not Docker-accessible
macOS XGBoost	Default install	brew install libomp — OpenMP required; libgomp1 in Docker
Airflow startup	Separate web-server/scheduler	airflow standalone — single command for Airflow 2.8
Pandas groupby	Standard apply	include_groups=False — pandas 2.2 deprecation fix
Port numbers	5000, 8000, 3000	5001, 8002, 3002 — host port conflicts resolved

12.1 Pickle Model Traceability

The bundled model maintains full traceability:

- `models/production_model.pkl` — the XGBoost model (v17)
- `models/production_model_meta.json` — embeds MLflow `version` and `run_id`
- `scripts/export_model_for_docker.py` — reproducible export script

13 Documentation Summary

Document	Status	Contents
docs/HLD.md	✓ Done	Architecture, technology choices, design rationale
docs/LLD.md	✓ Done	All 6 endpoint I/O specs with JSON examples
docs/test_plan.md	✓ Done	22 test cases with acceptance criteria
docs/test_report.md	✓ Done	22 passed, 0 failed, 88% coverage
docs/user_manual.md	✓ Done	5-screen guide for non-technical users
README.md	✓ Done	Full project setup, Docker, API reference, badges
Architecture Diagram	✓ Done	Added to HLD.md (ASCII block diagram)

14 Reproducibility — Setup & Run Guide

14.1 Prerequisites

- Conda (Python 3.10), Docker Desktop, Git LFS
- Kaggle API credentials in ~/.kaggle/kaggle.json

14.2 Full Docker Deployment (Recommended)

```

1 git clone https://github.com/DA5402-MLOps-JAN26/Final_Project_SPEWS_MA25M007.
  git
2 cd Final_Project_SPEWS_MA25M007
3 docker-compose up --build -d

```

Listing 5: Single-command deployment

Service	URL
Streamlit Dashboard	http://localhost:3002
FastAPI + Swagger UI	http://localhost:8002/docs
MLflow UI	http://localhost:5001
Airflow UI	http://localhost:8080 (admin/admin)
Grafana Dashboard	http://localhost:3001 (admin/admin)
Prometheus	http://localhost:9090

14.3 Generate Grafana Metrics

```

1  # Default: 400 requests, 180 s duration, dips 2-8 s, 20 spikes
2  bash populate_metrics.sh
3
4  # Custom requests and duration (random profiles over 3 min)
5  bash populate_metrics.sh 400 180
6
7  # Fine tune spikes and dips (30 spikes, long quiet periods)
8  bash populate_metrics.sh 400 300 5 15 30
9
10 # General form:
11 # bash populate_metrics.sh [REQUESTS] [DURATION_SEC] [MIN_DIP_SEC] [
    MAX_DIP_SEC] [BURSTS]

```

Listing 6: Populate metrics for Grafana panels – flexible bursty traffic

14.4 Run Tests

```

1  conda activate spews
2  pytest tests/ -v --cov=. --cov-report=term-missing
3  # Result: 22 passed, 88% coverage

```

Listing 7: Test suite with coverage

15 Conclusion

The **Student Performance Early Warning System** successfully demonstrates a production-grade MLOps pipeline. The system is notable for several technically mature design choices:

1. **Incremental retraining** — the model accumulates weekly data and is promoted only when F1 improves, demonstrating real model lifecycle management
2. **Drift detection with pipeline branching** — PSI score triggers Airflow retraining path automatically, not just an alert
3. **Full observability stack** — Prometheus scrapes API metrics every 15 seconds; Airflow pushes pipeline metrics via Pushgateway; Grafana provides a unified 6-panel NRT dashboard
4. **Strict loose coupling** — frontend and backend are truly independent containers connected only via configurable REST API calls
5. **Reproducibility** — every experiment is identifiable by Git commit hash + MLflow run ID; DVC tracks all data artifacts

Final Submission Status

Git Tags	v0.1-foundation → v0.2-ml-pipeline → v0.3-serving → v1.0-submission	✓
Tests	22/22 passed, 88% coverage	✓
Docker	All 7 services running via docker-compose up	✓
Model	F1=0.8651, AUC=0.9254 (targeted)	✓
Docs	HLD, LLD, Test Plan/Report, User Manual, README	✓